ON100B - SECTION 23

ECON100B - SECTION 23

International Trade and Exchange Rates

Felix Zhang

ANNOUNCEMENTS/AGENDA

Announcements

- PSET 4 released, due Friday 5/1

Agenda

- International Trade
- Exchange Rates

Question of the Day

Problem Set 4, Q1.a, Q1.b

(a) [2 points] Suppose that a nation engages in what becomes a protracted war overseas. But no military reservists are called up and away from their civilian jobs. What is the most obvious effect on that nation's GDP and why? (There's no need to get fancy with your answer.)

(b) [3 points] What, if anything, is the danger posed by financing the war through government budget deficits? Explain. (There are multiple correct answers; just explain one.)

International Trade

The growth of trade

- In the U.S., imports as a share of GDP have exceeded 15%, while they were only 4% in 1950

- Significantly reduced costs in shipping
- Significantly reduced costs in communication
- Reduced trade barriers

- Many countries import and export at a much higher share of GDP compared to the U.S.

Why should we trade?

- Forget the international space, just consider a domestic economy
- If we didn’t trade, we would all be responsible for things like sourcing/producing/making our own food, clothes, housing, etc.
- Trade allows for specialization, so people can do what they’re good at in exchange for other goods/money

- International trade follows this same logic, but just at a larger scale with greater specialization and knowledge domains
- International trade also helps with “risk-sharing”

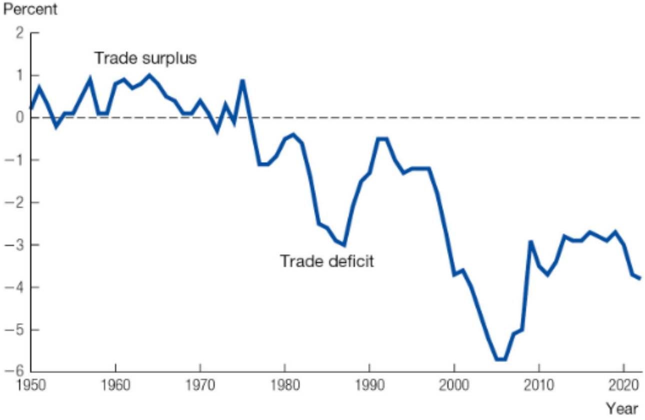
- Countries have comparative advantages in the production of certain goods and services and trade can make everyone better off

The Costs of Trade

- In the real world, trade isn't so perfect and can have negative impacts
- Under comparative advantage, we incentivize people in certain parts of the world towards certain jobs – but this can cause friction, job loss, and unemployment in certain sectors
- Some studies estimate that imports from China may have accounted for ¼ of the decline in U.S. manufacturing employment between 1990-2007
- Trade can lead to "winners" and "losers", especially when the benefits from trade aren't shared across everyone

The Trade Balance

The U.S. Trade Balance as a Share of GDP



Source: National Income and Product Accounts.

Trade deficits and surpluses

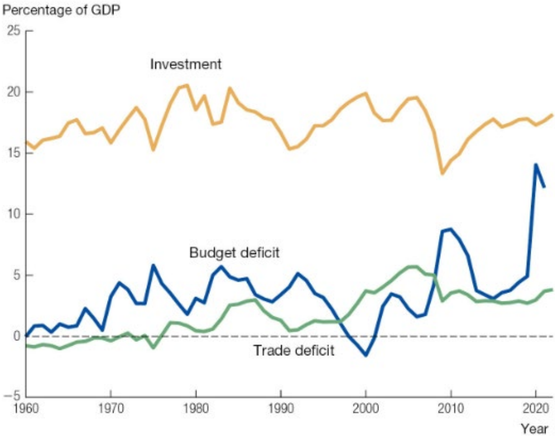
- Consider our manipulated national income identity

$$\underbrace{NX}_{\text{trade balance}} = \underbrace{S - I}_{\text{net capital outflow}} .$$

- The trade deficit can be understood as countries that invest more than they save
- The trade surplus can be understood as countries that save more than they invest
- For countries to run trade deficits, they need to provide something in return to countries that run surpluses
- When we buy goods from other countries with USD, they receive a financial instrument that they can invest in the U.S.

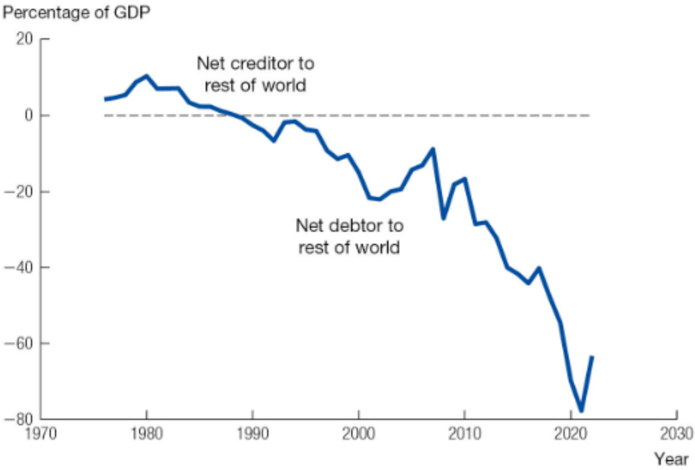
The Twin Deficit

U.S. Investment, Budget Deficit, and Trade Deficit



Source: National Income and Product Accounts.

Net International Investment Position of the United State

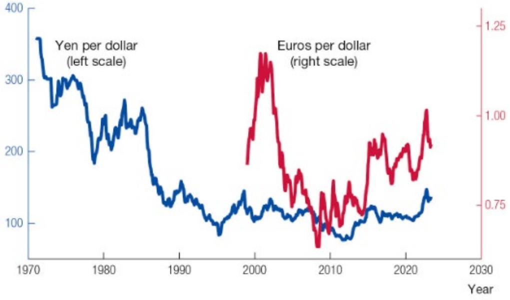


Exchange Rates

Exchange Rates

- The nominal exchange rate is the rate at which one currency trades for another, which we will denote with the letter “E”
- The “price” of a dollar has decreased over time, indicating depreciation

The U.S. Exchange Rate versus the Yen and the Euro



Source: The FRED database.

Long Run Exchange Rates

- In the long run, we expect the law of one price to hold. That is the value of the same good should be the same no matter where in the world
- We then relate exchange rates as the price level in one country to the price level domestically

$$\bar{E} = \frac{\bar{P}^{w}}{\bar{P}}.$$

- The long-run value of the nominal exchange rate can be understood through the quantity theory of money, or the relative nominal price of goods in two countries, i.e. the relative supplies of the two currencies

Short Run Exchange Rates

- People trade currencies for two reasons
1. Facilitate international trade
2. Financial markets demand currency to make financial transactions

- Let’s consider the scenario that we have a fixed supply of currency, set by the central bank
- What happens if interest rates in the U.S. increase? Why would people trade currencies and how would this influence the exchange rate?

PSET 4, Q 1.e

Politicians who like low inflation (π) typically also like a “strong dollar,” meaning that the dollar’s nominal exchange rate (E) is high, for example when the dollar ($) purchases many yen (¥) or euros (€).

(e) [4 points] State whether these two views are consistent or inconsistent with one another, and explain why. (There is a single correct answer, meaning either one or the other, but I think you could defend it at least two different ways. State at least one explanation.)

PSET 4, Q1.f

(f) [3 points] From the following list of constituencies (meaning political groups represented in the nation and in government), identify those that the "strong dollar" politicians probably represent, and briefly explain why.

- Exporters
- Consumers of imports
- Savers
- Borrowers
- Workers
- Retirees